

CS 189/289A: Homework 5

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Honor Code

“I certify that all solutions in this document are entirely my own and that I have not looked at anyone else’s solution. I have given credit to all external sources I consulted.”

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1 Self-Supervised Learning for Biology

(a) Interpreting the Objective

Minimizing the InfoNCE objective can be interpreted as an N -way classification problem because the loss function takes the exact form of the categorical cross-entropy loss applied to a softmax distribution over N candidates. For a given reference cell, the model is presented with N samples (1 positive and $N - 1$ negatives) and must output a probability distribution over these N choices.

The “classes” being predicted are the indices of the candidate samples in the given batch (from 1 to N). The ground truth class is the index corresponding to the positive sample z_i^+ .

(b) Gradient of the Contrastive Loss

Let $s_+ = (z_i^+)^T W c_i$ be the score for the positive sample, and $s_j = (z_j^-)^T W c_i$ be the score for the j -th negative sample. Let the denominator of the softmax be $Z = \exp(s_+) + \sum_{k=1}^{N-1} \exp(s_k)$. We can define the softmax probability for any sample k as $p_k = \frac{\exp(s_k)}{Z}$. The loss can be rewritten as $\mathcal{L}_i = -\log p_+ = -s_+ + \log Z$.

(i) **For the positive sample:**

$$\frac{\partial \mathcal{L}_i}{\partial s_+} = -1 + \frac{\exp(s_+)}{Z} = p_+ - 1$$

Since $p_+ \in (0, 1)$, this gradient is strictly negative. During gradient descent (subtracting the gradient), this encourages the network to increase the score s_+ , thereby pulling the latent representations of biologically similar cells closer together.

(ii) **For a negative sample j :**

$$\frac{\partial \mathcal{L}_i}{\partial s_j} = \frac{\exp(s_j)}{Z} = p_j$$

Since $p_j \in (0, 1)$, this gradient is strictly positive. Gradient descent will decrease the score s_j , effectively pushing the latent representations of unrelated cells apart.

(c) Effect of More Negative Samples

Mathematically, the InfoNCE loss provides a lower bound on the mutual information $I(Z; C)$ between the representations and the context: $I(Z; C) \geq \log(N) - \mathcal{L}_{\text{InfoNCE}}$. As N increases, the bound $\log(N)$ becomes larger, forcing the model to extract more distinctive features to identify the single positive sample among a massive pool of negatives, which leads to higher mutual information and better representation quality.

The tradeoff introduced during training is computational complexity and memory usage. Encoding and computing similarities for a much larger number of negative samples per reference cell requires significantly more memory and compute per batch.

(d) Contrastive Learning vs. Reconstruction

Directly reconstructing x_i forces the autoencoder to model all low-level, high-frequency details of the input, including stochastic noise, gene dropouts, and technical sequencing artifacts inherent to scRNA-seq. Contrastive learning in the latent space bypasses this by forcing the model to only encode the invariant, high-level semantic information (i.e., true biological identity) required to distinguish related cells from unrelated ones, ignoring irrelevant noise.

(e) Transfer Learning After Pretraining

This setup can outperform training from scratch because the self-supervised pretraining phase leverages massive amounts of unlabeled data to learn a rich, generalized, and robust feature space. Training a small classifier on top of frozen features requires optimizing far fewer parameters.

This is especially useful when labeled data is limited because deep supervised models easily overfit on small datasets. By pretraining, the model already understands the underlying structure of the data, meaning the small labeled dataset is only used to align those existing concepts to specific human-defined classes, drastically reducing the sample complexity required for the downstream task.

2 Diffusion Models

(a) Conditional Expectation

We want to find $f(y)$ that minimizes $\mathbb{E}_{x,y}[||f(y) - X||^2]$. By the Law of Total Expectation, we can condition on Y :

$$\mathbb{E}_{x,y}[||f(y) - X||^2] = \mathbb{E}_y [\mathbb{E}_{x|y} [||f(y) - X||^2 | Y = y]]$$

To minimize the overall expectation, we can minimize the inner expectation for each specific y . Taking the derivative of the inner expectation with respect to $f(y)$ and setting it to zero:

$$\frac{\partial}{\partial f(y)} \mathbb{E}_{x|y}[||f(y) - X||^2 | Y = y] = 2\mathbb{E}_{x|y}[f(y) - X | Y = y] = 0$$

$$f(y) - \mathbb{E}[X | Y = y] = 0 \implies f(y) = \mathbb{E}[X | Y = y]$$

(b) Optimal Denoising Function

We are given $L_\sigma(\epsilon_\sigma) = \mathbb{E}_{x_0, \epsilon} [||\epsilon_\sigma(x_0 + \sigma\epsilon) - \epsilon||^2]$, where $x_\sigma = x_0 + \sigma\epsilon$. We want to predict ϵ given the observation x_σ . Using the result from part (a) where $Y = x_\sigma$ and $X = \epsilon$, the optimal function is:

$$\epsilon_\sigma^*(x_\sigma) = \mathbb{E}[\epsilon | x_\sigma]$$

Since $\epsilon = \frac{x_\sigma - x_0}{\sigma}$, we can write:

$$\mathbb{E}[\epsilon | x_\sigma] = \mathbb{E}\left[\frac{x_\sigma - x_0}{\sigma} \middle| x_\sigma\right] = \frac{x_\sigma - \mathbb{E}[x_0 | x_\sigma]}{\sigma}$$

By Tweedie's formula, the posterior expectation of the clean data given the noisy observation is $\mathbb{E}[x_0 | x_\sigma] = x_\sigma + \sigma^2 \nabla \log p_\sigma(x_\sigma)$. Substituting this back yields:

$$\epsilon_\sigma^*(x_\sigma) = \frac{x_\sigma - (x_\sigma + \sigma^2 \nabla \log p_\sigma(x_\sigma))}{\sigma} = -\sigma \nabla \log p_\sigma(x_\sigma)$$

(c) Continuity Equation

Given the deterministic forward evolution $x_t = x_0 + \sigma(t)\epsilon$, the time derivative of a particle is $\dot{x}_t = \frac{d}{dt}(\sigma(t)\epsilon)$. The vector field is defined as the expected instantaneous velocity conditioned on the current state:

$$v(x, t) = \mathbb{E}\left[\frac{d}{dt}(\sigma(t)\epsilon) \middle| x_t = x\right] = \mathbb{E}[\dot{x}_t | x_t = x]$$

By the definition of the probability flow underlying a process, the marginal density $p_t(x)$ of particles evolving such that their expected velocity at position x and time t is $v(x, t)$ must conserve probability mass. The Eulerian perspective of mass conservation states that the rate of change of density at a point plus the divergence of the probability flux $v(x, t)p_t(x)$ must equal zero. Hence, the marginal density $p_t(x)$ exactly satisfies the continuity equation:

$$\partial_t p_t(x) + \nabla \cdot (v(x, t)p_t(x)) = 0$$

3 Flow Matching with General Interpolation Paths

(a) General velocity field

Given the time-dependent interpolation path $x_t = \alpha(t)x_1 + \beta(t)x_0$, we take the derivative with respect to t to find the ground-truth velocity:

$$\dot{x}_t = \frac{d}{dt} (\alpha(t)x_1 + \beta(t)x_0) = \alpha'(t)x_1 + \beta'(t)x_0$$

For the linear choice $\alpha(t) = t$ and $\beta(t) = 1 - t$, their derivatives are $\alpha'(t) = 1$ and $\beta'(t) = -1$. Substituting these into the velocity expression gives:

$$\dot{x}_t = (1)x_1 + (-1)x_0 = x_1 - x_0$$

which reduces exactly to the constant velocity.

(b) Single-function parameterization

(i) Velocity expression:

Given $x_t = h(t)x_1 + (1 - h(t))x_0$, taking the derivative with respect to time t :

$$\dot{x}_t = h'(t)x_1 - h'(t)x_0 = h'(t)(x_1 - x_0)$$

(ii) Effect of $h(t)$ on speed:

The magnitude of the velocity vector is $\|\dot{x}_t\| = |h'(t)| \cdot \|x_1 - x_0\|$. Since $h(t)$ is monotonically increasing, $h'(t) \geq 0$. The term $h'(t)$ acts as a scalar multiplier that strictly determines the instantaneous speed along the linear trajectory from x_0 to x_1 . A larger $h'(t)$ means the path moves faster at time t .

(iii) Comparing choices of $h(t)$:

- $h(t) = t$: The derivative is $h'(t) = 1$. The speed is strictly constant throughout the entire trajectory.
- $h(t) = t^2$: The derivative is $h'(t) = 2t$. At $t = 0$, $h'(0) = 0$, and at $t = 1$, $h'(1) = 2$. The trajectory moves slowly at the beginning and accelerates, moving **faster near** $t = 1$.
- $h(t) = \sin(\frac{\pi t}{2})$: The derivative is $h'(t) = \frac{\pi}{2} \cos(\frac{\pi t}{2})$. At $t = 0$, $h'(0) = \frac{\pi}{2}$, and at $t = 1$, $h'(1) = 0$. The trajectory moves quickly at the beginning and decelerates, moving **faster near** $t = 0$.